OS skill-21

To Connect Modules, Validate Interfaces, Test End-to-End Execution, Handle Errors, Improve Reliability, Refactor Components.

To Detect Runtime Errors, Generate Messages, Log Failures, Handle Invalid Syntax, Recover Gracefully, Test Error Scenarios.

nano calculator.h

```
#ifndef CALCULATOR_H
#define CALCULATOR_H

int add(int a, int b);
int multiply(int a, int b);

#endif
```

nano calculator.c

```
#include "calculator.h"

int add(int a, int b) {
return a + b;
}

int multiply(int a, int b) {
return a * b;
}
```

nano main.c

```
#include <stdio.h>
#include "calculator.h"

int main() {
int a = 10, b = 5;

printf("Addition = %d\n", add(a, b));

printf("Multiplication = %d\n", multiply(a, b));


return 0;
```

}

```
lahari@DESKTOP-AIC33F7:~$ mkdir module_testing
cd module_testing
lahari@DESKTOP-AIC33F7:~/module_testing$ nano calculator.h
lahari@DESKTOP-AIC33F7:~/module_testing$ nano calculator.c
lahari@DESKTOP-AIC33F7:~/module_testing$ nano main.c
lahari@DESKTOP-AIC33F7:~/module_testing$ gcc -Wall -Wextra main.c calculator.c -o app
lahari@DESKTOP-AIC33F7:~/module_testing$ ./app
Addition = 15
Multiplication = 50
lahari@DESKTOP-AIC33F7:~/module_testing$ gcc -Wall -Wextra -Werror main.c calculator.c -o app
./app
Addition = 15
Multiplication = 50
lahari@DESKTOP-AIC33F7:~/module_testing$ ~
```

nano error_handling.c

```
#include <stdio.h>
int main() {
int a, b;
FILE *log = fopen("error.log", "a");

printf("Enter two integers: ");

if (scanf("%d %d", &a, &b) != 2) {
   printf("Error: Invalid input syntax.\n");

   if (log != NULL) {
      fprintf(log, "Error: Invalid input syntax.\n");
      fclose(log);
   }

   return 1;
}

if (b == 0) {
   printf("Error: Division by zero is not allowed.\n");

   if (log != NULL) {
      fprintf(log, "Error: Division by zero.\n");
      fclose(log);
   }

   return 1;
}
```

```
printf("Result = %.2f\n", (double)a / b);

if (log != NULL) {
   fprintf(log, "Program executed successfully.\n");
   fclose(log);
}

return 0;
}
```

```
lahari@DESKTOP-AIC33F7:~/module_testing$ nano error_handling.c
lahari@DESKTOP-AIC33F7:~/module_testing$ gcc -Wall -Wextra error_handling.c -o error_handling
lahari@DESKTOP-AIC33F7:~/module_testing$ ./error_handling
Enter two integers: 10 2
Result = 5.00
lahari@DESKTOP-AIC33F7:~/module_testing$ ./error_handling
Enter two integers: 10 0
Error: Division by zero is not allowed.
lahari@DESKTOP-AIC33F7:~/module_testing$ ./error_handling
Enter two integers: hello 5
Error: Invalid input syntax.
lahari@DESKTOP-AIC33F7:~/module_testing$ cat error.log
Program executed successfully.
Error: Division by zero.
Error: Invalid input syntax.
lahari@DESKTOP-AIC33F7:~/module_testing$ ./error_handling
Enter two integers: 20 4
Result = 5.00
lahari@DESKTOP-AIC33F7:~/module_testing$
```